

Quiz #7

CS 225 Data Structures

Spring 2026

7:00-8:00pm, Thursday, May 21

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Lab TA	

- This is a **closed book** and **closed notes** exam. No electronic aids are allowed, either.
- You should have 4 problems total on 4 pages. You must write on the exam.
- Unless otherwise stated in a problem, assume the best possible design of a particular implementation is being used.
- Unless the problem specifically says otherwise, (1) assume the code compiles, and thus any compiler error is an exam typo (though hopefully there are not any typos), and (2) assume you are NOT allowed to write any helper methods to help solve the problem, nor are you allowed to use additional arrays, lists, or other collection data structures unless we have said you can.
- Please put your name at the top of each page.
- Please fill in your **Lab Section** or your **Lab TA's name** to help us organize and grade your papers.
- Don't panic and good luck!

Problem 1 _____/10

Problem 2 _____/10

Problem 3 _____/10

Problem 4 _____/10

Total _____/40

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Problem 1 (10 points): Graph Implementations and Runtime

Assume the graph has n vertices and m edges. Unless otherwise stated, assume the best possible implementation is used.

1. Suppose a graph is implemented using an edge-list. What is the runtime to check whether there is an edge between two given vertices?

- a) $O(1)$
- b) $O(\log n)$
- c) $O(n)$
- d) $O(m)$
- e) $O(n + m)$

2. What is the main advantage of using an adjacency matrix?

- a) It uses less space for sparse graphs
- b) It can check whether an edge exists between two vertices in $O(1)$ time
- c) It makes BFS always $O(n + m)$
- d) It is always faster than adjacency list
- e) None of above are correct

3. What is the runtime of BFS using an adjacency matrix and adjacency list, respectively?

- a) $O(n + m)$, $O(n^2)$
- b) $O(n^2)$, $O(n + m)$
- c) $O(m)$, $O(n)$
- d) $O(n \log n)$, $O(m \log n)$
- e) $O(n^2)$, $O(n^2)$

$$O(n + 2m)$$

for $(V \text{ Vertices})$
BFS(u)

4. What is the runtime of removeVertex in an adjacency matrix representation?

- a) $O(1)$
- b) $O(\log n)$
- c) $O(n)$
- d) $O(n \log n)$
- e) $O(n^2)$

	0	1	2	3
0				
1				
2				
3				

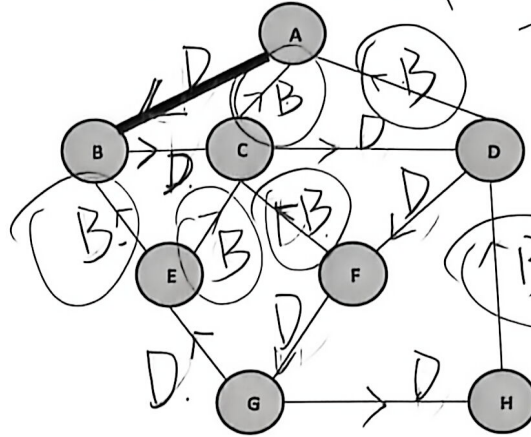
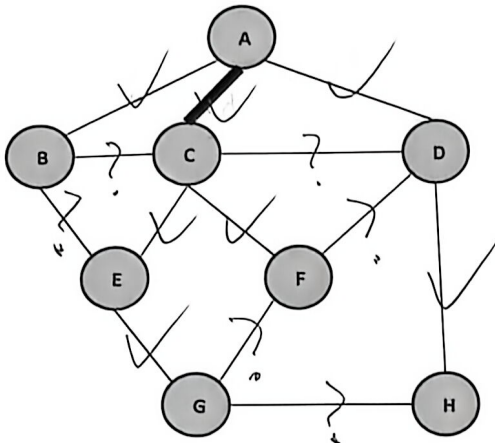
5. Which statement is correct about graph algorithms and runtime?

- a) Dijkstra's Algorithm works correctly even with negative edge weights
- b) Floyd-Warshall finds all-pairs shortest paths in $O(n^3)$ time
- c) Prim's Algorithm is used to find single-source shortest paths
- d) Kruskal's Algorithm only works on directed graphs
- e) None of above are correct



Name: ✓

Problem 2 (10 points): iterator & recursion



1. Apply **BFS** graph traversal algorithm to the above graph starting from vertex A. Specifically, draw all **DISCOVERY** edges in the picture above as you follow the BFS traversal and fill in the metadata table below with distance (d) and predecessor (p). **Unlike what's given in the lecture, First record and edge** are already provided for you to follow with your BFS search. For rest of the case you should use alphabetic order to choose the next vertex to search from the adjacent vertices. Use as many rows as needed.

d	p		Adjacent Vertices
0	A	A	C B D

2. Apply **DFS** graph traversal algorithm to the above graph starting from vertex A. Specifically, draw all **DISCOVERY** edges in the picture above as you follow DFS algorithm and then list all **DISCOVERY** and **BACK** edges below. You can identify edges by the two vertices they connect. First edge is already provided for you to follow with your DFS search. You should visit the **next** vertices in **alphabetical order** when multiple candidates are available.

DISCOVERY edges:

AB BC CD DF
FG GE GH

BACK edges:

EC EB EC HD
DA

for v vertex 6



Problem 3 (10 points): Shortest Path Algorithms

1. In an unweighted graph, BFS starting from a source vertex s guarantees which of the following?

- a) Vertices are first discovered in nondecreasing shortest-path distance from s , where distance is measured by the number of edges.
- b) The algorithm always returns a minimum spanning tree of the graph.
- c) The algorithm is correct only when the graph is directed and acyclic.
- d) The algorithm computes shortest-path distances only when all edge weights are distinct.
- e) None of the above.

2. Consider standard Dijkstra's algorithm on a directed graph whose edge weights are all nonnegative. When a reachable vertex m is removed from the min-priority queue, which statement is guaranteed to be true?

- a) $d[m]$ may still decrease later in the algorithm.
- b) $p[m]$ must be NULL.
- c) $d[m]$ equals the length of a shortest path from the source s to m .
- d) m is necessarily unreachable from s .
- e) The algorithm must terminate immediately after removing m .

3. Which statement about the Floyd-Warshall algorithm is correct?

- a) It computes shortest paths only from one specified source vertex.
- b) It requires every edge weight to be nonnegative.
- c) It computes all-pairs shortest-path distances and can handle negative-weight edges as long as the graph has no negative-weight cycle.
- d) It is equivalent to BFS on every weighted graph.
- e) None of the above.

4. Suppose standard Dijkstra relaxation is applied to a directed edge (u, v) . Immediately before the check, $d[u] = 6$, $\text{cost}(u, v) = 5$, $d[v] = 14$, and $p[v]$ is arbitrary. After processing this edge, what are the updated values?

Answer: $d[v] = 11$; $p[v] = u$

5. Standard Dijkstra's algorithm, as presented in class, is guaranteed to be correct only when every edge weight in the graph is non-negative.

Answer: _____



Problem 4 (10 points): MST & Learning by Doing

For each MCQ below, there is only exactly one correct answer.

1. Which statement is true about Kruskal's and Prim's algorithms? (MST)

- ~~a) Both algorithms find shortest paths from a source vertex.~~
 b) Kruskal grows the MST by choosing globally smallest safe edges, while Prim grows from a starting vertex.
 c) Prim requires Disjoint Sets, while Kruskal does not.
 d) Kruskal cannot be used on undirected graphs.

2. What is the runtime of Kruskal's Algorithm if the priority queue is implemented using a heap? Assume the graph has n vertices and m edges.

- a) $O(n + m)$
 b) $O(n + m \log n)$
 c) $O(n^2)$
 d) $O(m^2)$
 e) $O(n \log m)$

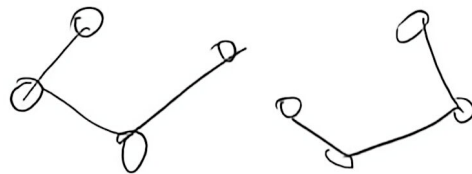
$$O(n) + O(m) + O(m \log n)$$

disjoint set () $O(n)$
 pq build $O(m)$

$$n + m + m \cdot \log m$$

3. Which of the following must be true for a Minimum Spanning Tree of a connected undirected graph with n vertices?

- a) It contains exactly n edges.
 b) It contains exactly $n - 1$ edges.
 c) It may contain cycles if all edge weights are positive.
 d) It is always unique.



4. Briefly discuss two things that you learned by completing any of your lab or mp assignments. In each case, describe what you learned and how you learned it (2 point) and how you will apply this lesson to your future projects (2 point). You can talk about tools, design, working with your partner, time management — whatever you want, as long as it is something that you learned this semester from CS 225 lab assignments.

A)

B)

